

Scan2Fab AI Pipeline

Instant NeRF-to-CNC Reverse Engineering Workflow

0%

HARDWARE COST



PHASE 1: CAPTURE

Physical Edge

An unskilled MSME worker records a 10-second orbital video of a broken mechanical part using any standard smartphone camera.

- ✓ Zero Capital Cost
- ✓ No Skill Required



PHASE 2: COMPUTE

NeRF Cloud Engine

The cloud AI utilizes **Neural Radiance Fields (NeRF)** and Gaussian Splatting to extrapolate 3D depth and geometry from 2D video frames.

- ⚡ Generative Spatial AI
- ⚡ Sub-millimeter Accuracy



PHASE 3: SYNTHESIZE

Digital Twin Mesh

The AI instantly generates a mathematically perfect 3D digital mesh, exporting it as an



PHASE 4: EXECUTE

Factory Machining

The CAD file is fed directly into the MSME's existing CNC machines or 3D printers to fabricate the physical replacement part instantly.

- ⚡ Zero Setup Time
- ⚡ Massive ROI